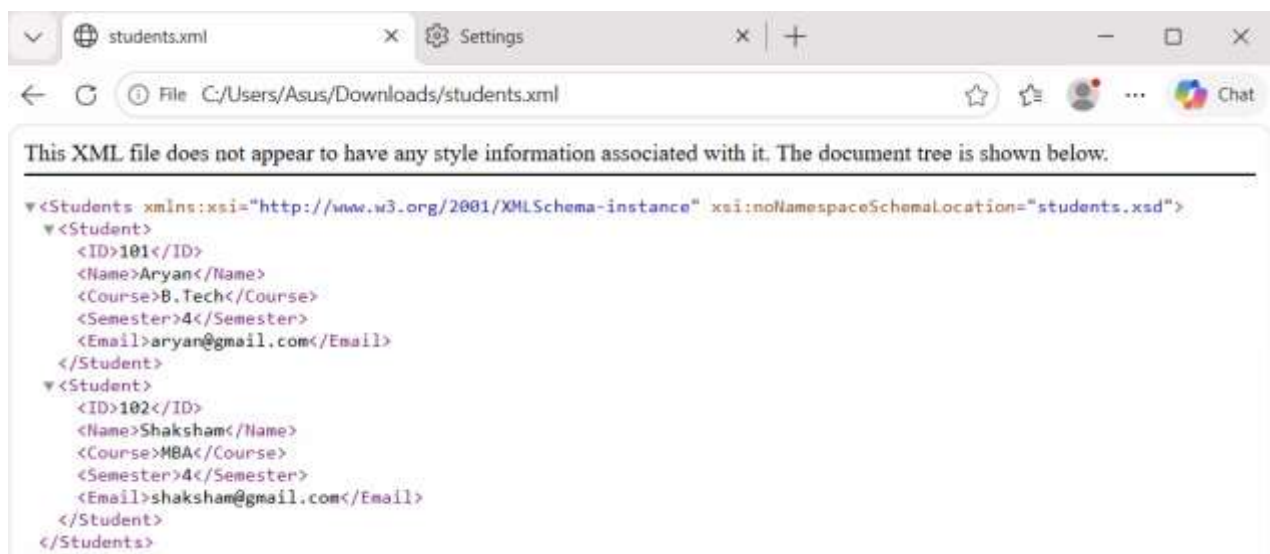


XML:



XSD:



Index: create index() & getindex()

```
> db.students.createIndex({name: 1})
< name_1
> db.students.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { name: 1 }, name: 'name_1' }
]
```

dropindex()

```
> db.students.dropIndex("name_1")
< { nIndexesWas: 2, ok: 1 }
```

```

> use libraryDB
< switched to db libraryDB
> use salesDB
< switched to db salesDB
>
  use studentDB
< switched to db studentDB

```

```

use LibraryDB
switched to db LibraryDB
db.books.insert([
{
  title: "Atomic Habits",
  author: "James Clear",
  year: 2018,
  available: true
},
{
  title: "The Alchemist",
  author: "Paulo Coelho",
  year: 1988,
  available: true
},
{

```

Create

```

acknowledged: true,
insertedIds: {
  '0': ObjectId('6a03f8b564441bf482cfb713'),
  '1': ObjectId('6a03f8b564441bf482cfb714'),
  '2': ObjectId('6a03f8b564441bf482cfb715')
}
}

```

Update & Delete

```

db.books.updateOne(
  {title: "Ikigai"},
  {$set: {available: false}}
)

db.books.deleteOne(
  {title: "Rich Dad Poor Dad"}
)
< {
  acknowledged: true,
  deletedCount: 1
}

```

Read

```

> db.books.find()
< {
  _id: ObjectId('69fc6324a1390e2e437a8b78')
}
{
  _id: ObjectId('6a03f96464441bf482cfb732'),
  title: 'Atomic Habits',
  author: 'James Clear',
  year: 2018,
  available: true
}

```

```
> db.sales.insertMany([
  {
    product: "Laptop",
    category: "Electronics",
    amount: 100000
  },
  {
```

```
    _id: 'Electronics',
    totalSales: 101000
  },
  {
    _id: 'Clothing',
    totalSales: 6500
  }
])
```

Aggregation

```
> db.sales.aggregate([
  {
    $group: {
      _id: "$category",
      totalSales: {$sum: "$amount"}
    }
  },
  {
    $sort: {totalSales: -1}
  }
])
< {
  _id: 'Electronics',
  totalSales: 101000
}
{
  _id: 'Clothing',
  totalSales: 6500
}
```